

**Trouble Shooting Laser
Problems Dimension**

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Products: **D3000, 3100, D5000, and V**

Revision	Description	Date
A	Initial Release – (Teddy Huang and Pat McPhail)	12/23/2014
B	Rev B – (Teddy Huang and Pat McPhail)	03/30/2015

Subject: (Trouble Shooting Laser Problems_Dimension)**System(s) Affected: (D3100, D5000, and DimV)****Action Category: (As Needed)****Summary: (Troubleshooting guide)****WARNING AND CAUTIONS:**

- The troubleshooting guide assumes the engineer is familiar working with a digital voltmeter and an oscilloscope on live circuits.
- When operating the electronic parts or components, use extreme caution and do not drop any metal objects into them that may lead to short-circuits.
- Do not reach into any electronic part or component while it is powered or connected to mains. Circuits carry **high-voltage**.
- Use electrostatic discharge protection, e.g. wrist strap, when handling electronics and printed circuit boards. Fine probes of digital voltmeter are required to avoid harmful short circuit.
- Measure the signals vs. GND and only make or break connections with the controller powered off.
- Do not tear at the test leads when connected to the (surface-mounted-device) SMD test point to prevent the SMD test points from detaching from the PCB.
- Only use isolated flexible SMD test clamps when measuring high-voltage signals. Risk of electrical shock, short-circuit and damage to electronic components.
- If there is any uncertainty, stop immediately and call Bruker Tech Support (+1-800-873-9750).

Pre-start reading:

- Before proceeding to diagnostic, check all cables are correctly connected and well-seated. A loose/half-plugged cable can cause “mysterious” issues. In addition, to proceed, Nanoscope software is required to be correctly configured according to the existing hardware. Pay attention to intermittent problems. They are quite likely caused by bad cables.
- If an extender exists, make sure the configuration is correct. Refer to support notes: “230A-EFM Dimension”, and “013-322-000, QUADREX PHASE IMAGING SUPPORT NOTE-D”.

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- When Nanoscope controller of NS3 series with extender used, make sure the correct motor chip is used. Please call Bruker Tech Support (+1-800-873-9750) if you are not clear.
- Make sure that the software is configured properly according to the existing hardware.

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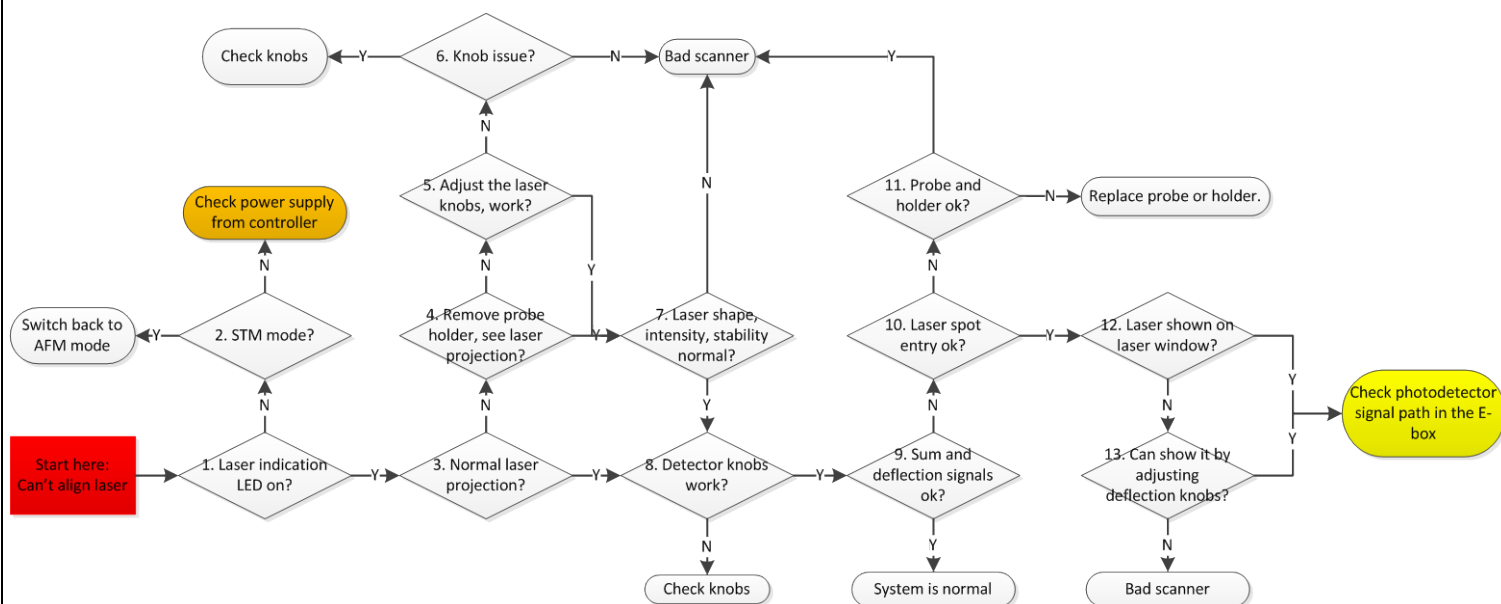
General logics:

Laser on. → Laser is not block by internal optics. → Laser can be aligned by screws. → Detector signals can be routed to electronic box (E-box). → E-Box can process signals into SUM and deflection. → SUM and deflection signals can be sent to controller. → Controller and DSP can process the signals. → SUM and Deflection signals can be shown on software.



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Section 1: Start here: Can't align the laser.



Step 1: Is the LED on the front of the scanner head on? See Figure 1.



Figure 1. Illustration of the front LED and the laser window on the front of the scanner head. This picture shows a normal working head with laser position and deflection properly aligned where the LED is lit and the laser spot on the laser window is at the center.

Step 2: When the STM mode on the software is enabled, the laser will be turned off and so the indication LED in Figure 1 will be off.

Addition note: if the AFM is NOT on STM mode, while the laser is off, one possibility is that STM is not following the software command and the STM control is always at Logic high, see appendix for

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measurement STM logic control. This could be due to corrupted Nanoscope software. If re-installation of the software does not help, the DSP could be bad.

Step 3: Check that the laser projection can be seen on a piece of white paper as shown in Figure 2. Check that the laser projection has no intermittency, no distorted shape and no dim brightness. The laser intensity can be checked by reading the sum signal.

There is a mercury switch inside the scanner. When the scanner is slightly tilted, the laser will be turned off. For this step, make sure the scanner is hold vertically. A good practice would be keeping the scanner in the dovetail, lift the scanner all way up with the software and put a piece of white paper underneath. Move the stage to the front left corner is possible. This requires a sufficient sum signal detected. Alternatively, the stage can be move by manually rotating the rubber grips on the motors, see Figure 3.

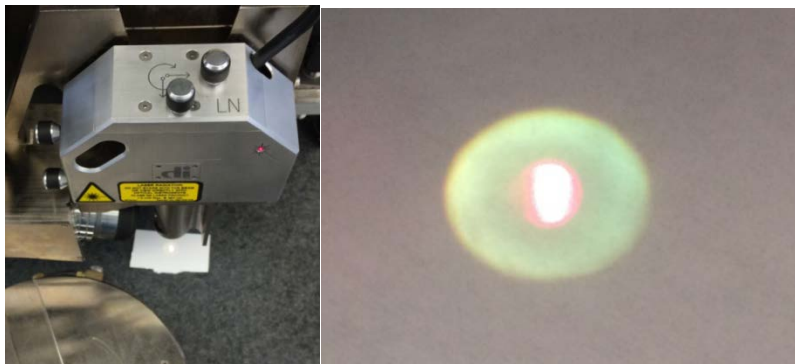


Figure 2. Illustration of laser projection on a piece of paper. The chuck stage is moved to the front left.

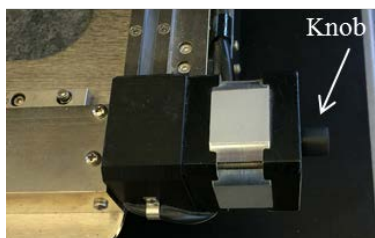


Figure 3. Rubber grips on the motor for manually moving the chuck stage.

Step 4: Remove the probe holder to check if the laser projection can be shown on a piece of white paper. The AFM probe or even the probe holder can block the laser if the laser position is not correctly adjusted.

Step 5: Turn laser screws clockwise all the way and then 2 turns back. You should be able to see the laser projection. If not, the laser is blocked inside the scanner.

Step 6: If the mechanical function of the knobs are in doubt (stuck or stripped), see Section 2.

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Step 7: See Step 3.

Step 8: After move the laser to the cantilever, firstly check the detector screws turned normally. If not, check knobs. See Section 2.

Step 9: For a normally working system, when adjusting the laser position and deflection knobs, the photodetector and the software should response correctly.

- (1) When unplug the head, on the software, all signal readings (vertical deflection, horizontal deflection and signal sum) are gone. The red dot on the quadrant meter on the software turns to pale grey.
- (2) When adjusting the laser position screws on the top of the scanner head, after the laser spot hits the cantilever, the laser sum can be adjusted to a correct value. If you are not sure about what is the correct sum, contact Bruker Tech Support (+1-800-873-9750, AFMsupport@bruker-nano.com)
- (3) When adjusting the laser deflection screws on the left of the scanner head, the vertical deflection, horizontal deflection and signal sum can be adjusted accordingly.
- (4) The deflection signals can swipe between ± 10 V showing on the software and being simultaneously illustrated by the moving red dot on the software meter. Meanwhile, the laser spot on the laser window of the head move accordingly.
- (5) When swiping the deflection signal only on one direction, the red dot on the quadrant meter on the software is not moving diagonally.
- (6) When the sum is maximized and the laser deflections are adjust to zero, the laser spot on the laser window is approximately at the center and the dot on the software meter is red and at the center.

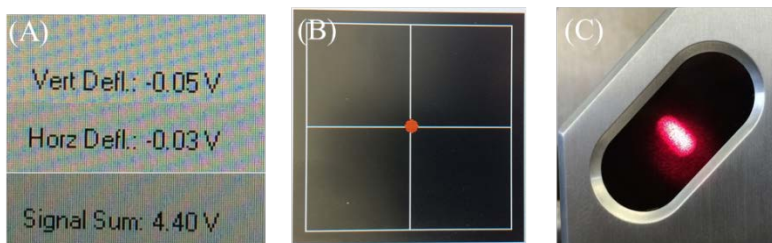


Figure 4. Correct response of photodetector and software after the laser is correctly aligned. Note that the Signal Sum value depends on a specific probe. (A) Software readings of vertical deflection (Vert Defl.), horizontal deflection (Horz Defl.) and signal sum; (B) The quadrant meter. The red dot turns to pale grey if the head is unplugged; (C) Laser spot on the laser window of the head.

Step 10: If the probe and the probe holder are correctly seated, the laser will be reflected into the head through the laser spot entry window (Figure 5).

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1. Check that the window is clean. If not, clean by Kim Wipe moistened with IPA.
2. Use a piece of white paper to check that the laser is physically hit the entry window. While doing this, make sure the head is hold vertically. This is a two-person job for not dropping the head.

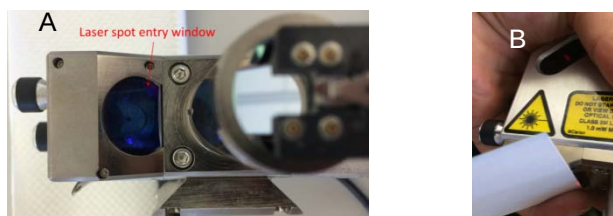


Figure 5. Illustration of the laser spot entry window. (A) Bottom view of the head with the entry window specified; (B) A business card is used for checking the laser is physically hit the entry window.

Step 11: First, check that the probe and the probe holder are well seated and mounted. If the problem remains, try another probe from another box of tip. If new probe does not work, try another probe holder that was confirmed good before. With a good probe and a good probe holder, if the problem remains, the head assembly is bad, resulting in an incorrect laser reflection angle.

Step 12-13: See step 9.

Section 2: Check Laser and Detector Alignment Knobs

For a working scanner head, the two knobs on the top can be turned smoothly and the laser spot move accordingly. The two knobs on the left can be tuned smoothly and the laser deflection move accordingly. If the probe holder mounted correctly, a normal probe (SNL, DNP, TEPSA, etc) loaded correctly and the screw threads has no problem (no stuck or strip), the laser should be able to move to the cantilever and allows deflection adjustment with no problem.

Screws can get stuck and prohibit for turning. Screw threads can also strip. If the screw does not stuck and the screw threads show no wearing out, however, the laser projection does not move or the laser spot on the laser window (Figure 4C) does not move, it is likely that stripping occurs inside the thread housing on the scanner. If this is the case, the head is required for factory repair. If removing screws is required, label different screws at different screw housings, as these four screws have different lengths, as shown in Figure 6. **If re-installed a screw at an incorrect slot, the adjustment will not work correctly and PCBs inside the scanner head can be damaged.**

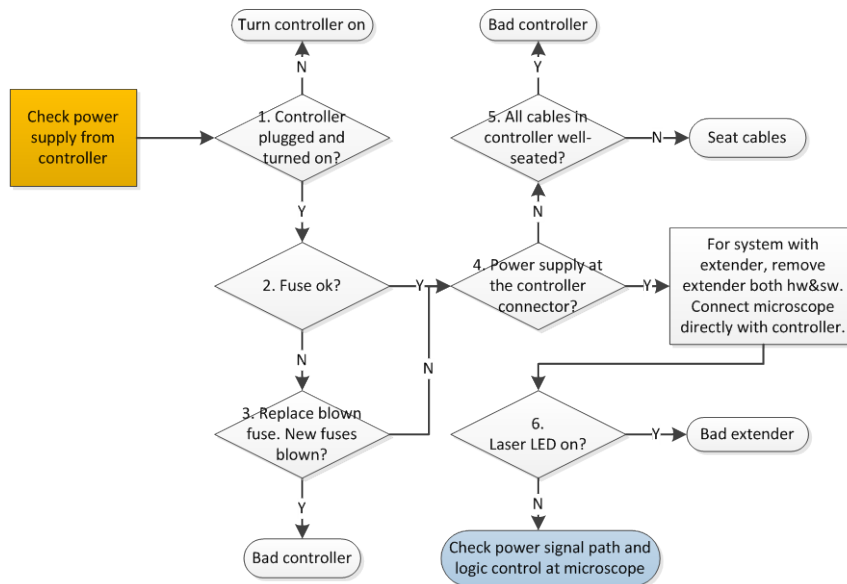
If screws get stuck, remove the screw and clean it with acetone, apply multiple purpose lithium grease to it and reattach the screw.

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Figure 6. The four screws for laser alignment. Be aware of the difference in length.

Section 3: Check power supply from controller



Step 1: Make sure the controller is plugged and switched on.

Step 2: Check fuses. Fuses can be access from the back of the controller. Figure 7 shows fuse housings on two different type of NSIII (or IIIA) controller. Figure 9A is a newer model than Figure 9B. For NSIV, IVA and V, they have the same fuse housing. For Figure 9A, first, check that the voltage selections match the line power. If not, use a quarter coin to adjust it. If fuses are blown, put new ones. If new fuses are blown, electrical short occur inside the controller.

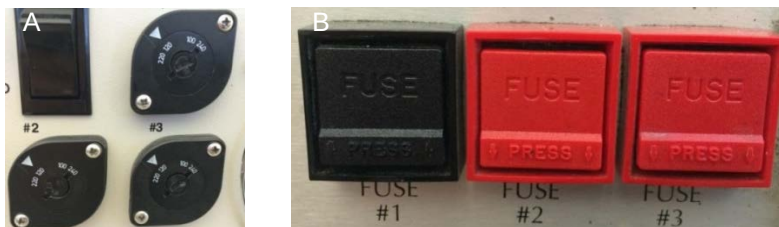


Figure 9. Fuse housings at the back of the controller. (A) A newer controller model; (B) An older model.

Step 3: If the new replaced fused keep blown, it is a bad controller.

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Step 4: Turn off the controller. Unplug the 37-pin cable that connecting the controller and the microscope. Turn on the controller. For non NSV controllers, check the 5V supply at Pin 20, +15 V at Pin 30 and -15V at Pin 9 on the 37-pin cable female connector (Figure 10(A)). A paper clip can help to get access into the pinhole for measurement. For NSV controller, see Figure 10(B) for test points.

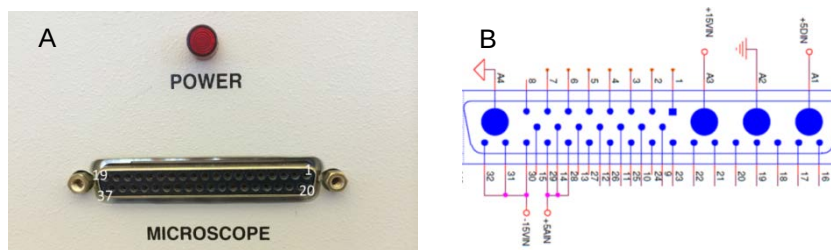


Figure 10. (A) Illustration of the power LED and 37-pin cable connector on non-NSV controller. The pin-out numbers are labelled. This connector is either in the front or in the back of the controller. (B) Pin out for High Voltage/Digital cable and connector of NSV controller.

Step 5: If the 5V and +/-15V are missing at the cable connector, open the controller to check if loose cables exist. If fuses are good and cables are all well-seated, while powers are missing at the front connector, the controller is required factory evaluation and repair.

We do have Tech Support Bulletin-NBSB-0008 for more details regarding NSIII series.

For NSIV series, open the top cover, remove the metal cover, the top layer is shown on Figure 11A. Remove the locking screws, the top layer can be lifted up and the bottom architecture is shown in Figure 11B. Tip: lift the top layer from the gap in front of the power assembly. The bottom PCB assemblies are the same as NSIII series and the Tech Support Bulletin-NBSB-0008 is also applicable.

For a NSV controller, refer to “NBSB0056 Testing Power Supply on NS5 and BIOII Controllers”

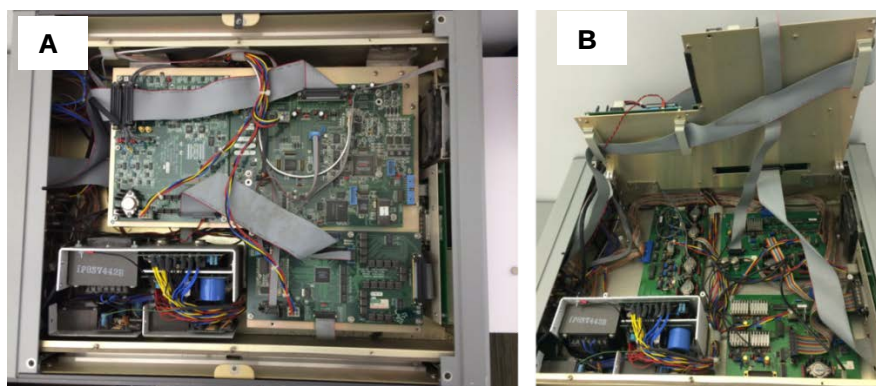
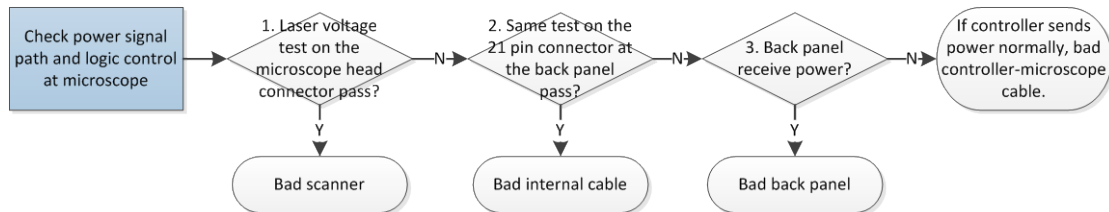


Figure 11. Illustration of the architectures inside a NSIV controller. (A) top layer; and (B) bottom layer.

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Step 6: This step is only for system with an extender. If no extender, skip this step. Remove the extender and change the software to the right configuration, connecting controller directly with the microscope.

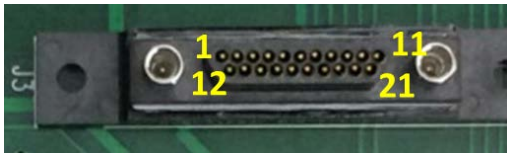
Section 4: Check power signal path and logic control in E-box



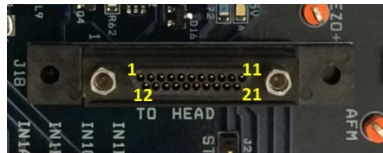
After opening the E-box, check all cables are well-seated and no visible damage, such as burnt holes or mechanical failures.

Step 1: Refer to “NBSB0071_Testing Laser Voltage from Back Panel”.

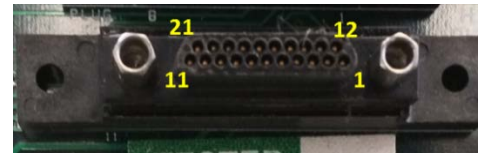
Step 2: Refer to “NBSB0071_Testing Laser Voltage from Back Panel”, but test on the cable connector on the e-box as shown in Figure 12.



(A)



(B)



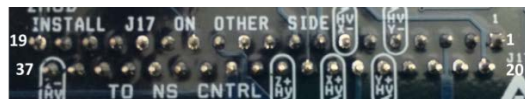
(C)

Figure 12. 25 pin scanner head connector on the back panel inside the electronic box. (A) PCB #312-000-005, for D3000 and D3100 systems; (B) PCB #312-000-181, for D3000 and D3100 systems; and (C) for Dimension V system.

Step 3: Measure low voltage supplies. For D3000 and D3100: +5V (Pin 20) , +15V (Pin 30) and -15 V (Pin 9), see Figure 13 (A) and Figure 13(B). For Dimension V: +5V (Pin 14, 28 or 29), +15V (A3) and -15 V (Pin 30, 31 or 32), see Figure 13 (C)



(A)



(B)

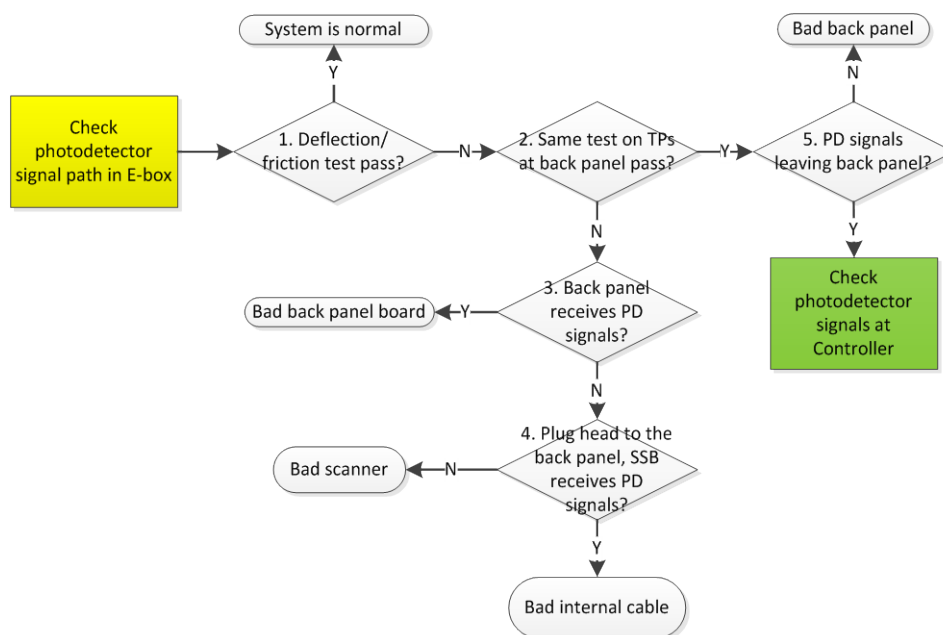


(C)

Figure 13. 37 pin cable connector (D3000, D3100, and D5000) and High Voltage connector (DimV) at the back panel in the electronic box. (A) PCB #312-000-005, for D3000 and D3100 systems. ; (B) PCB #312-000-181, for D3000 and D3100 systems; and (C) for Dimension V system.



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Section 5: Check photodetector signals in E-box

Step 1: Refer to “NBSB0005_Deflection and Friction Test” and “NBSB0021_Deflection and Friction Test v7”

Step 2-5: Do the deflection/friction test on the test points in the E-box. Refer to “NBSB0005_Deflection and Friction Test” and “NBSB0021_Deflection and Friction Test v7”

For NSV system: Photodetector signals from the four quadrant of the detector first arrive at the connector then pass to a set of resistors to generate corresponding vertical deflection, horizontal deflection and SUM signals. Quadrant A is R7, R14, and R33; Quadrant B is R5, R15, and R35; Quadrant C is R4, R17, and R34; and Quadrant D is R3, R19, and R37. Signals at {R3, R4, R5 and R7} are for lateral deflection signal; {R14, R15, R17 and R19} are for SUM signals; {R33, R34, R35, and R37} are for vertical deflection signal. These resistors are circled out in Figure 14.

For a working system, the raw photodetector signals can be measured at the legs of these resistors. Note that on each resistor, voltage signals are different from one leg to the other due to the voltage drop. After plugging the head and align the laser to the cantilever, if no reading from the legs of these resistors, the problem is either the head or a bad cable. To rule out the cable issue, the head can be directly plug into the cable connector on the back panel and perform the same test. The head need to be held vertically to avoid the mercury switch turning off the laser. This is a two-person job.

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For a working system, the reading on a resistor leg that for deflection signal by swiping the red dot diagonally is 0 to ~3V. See the table below for quadrant assignment. If you align the deflection to the center of the meter on the software, the SUM signal reading from each leg of the corresponding resistor is approximately a quarter of the total SUM. Again, be aware of the voltage drop across the resistor.

The raw photodetector signals are processed into SUM and deflection signals with corresponding test points for them as circled out in Figure 14. See Table 2 for test point functions. Keep in mind that missing signals at this test point is not necessarily to conclude a bad canner. It could be a bad cable or a bad processing circuit block on the head module board.

Table 1. Assignment of quadrant on the software meter.

A	B
C	D

Table 2: Functions of test points on SSB head module board.

Test Point	Reference	Function
TP1	IN1B	LP filtered vertical deflection
TP3	IN0	Vertical signal (deflection, tapping)
TP4	SUM	Laser sum signal
TP7	IN1A	Lateral signal (LFM, torsion)

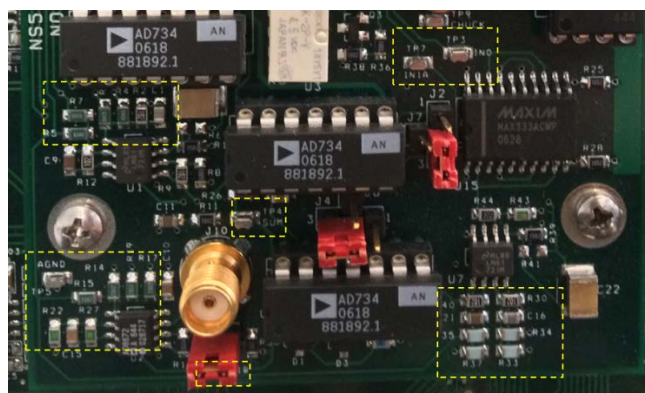


Figure 14. NS5, SSB Head Module Board with test points circled out.

The last place when these SUM and deflection signals leave the back panel is at the low voltage D78 cable connector at the back panel, as shown Figure 15. Note that signals arrived here have been processed by the back panel and been differentiated into two signal streams while the readings are the same. If the

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SUM and deflection signals read normally at the test points while missing at the corresponding pins at the D78 connector, it concludes a bad back panel. If signals read normally here, problems are at downstream.

A

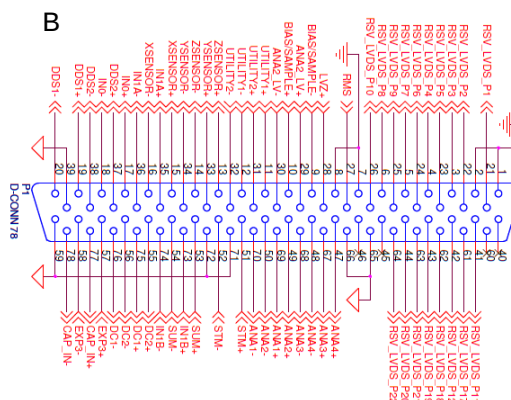


Figure 15. (A) The D78 low voltage cable connector at the back panel in the Ebox; (B) Pin out for the D78 connector

For D3000, D3100 with blue PCB #312-000-181: Refer to “**For NSV**”. See Table 3-4, Figure 16-17.

Table 3. Location for test photodetector signals, refer to Figure 7

Quadrant	Vertical	SUM	Horizontal
PDA	R111	R117	R131
PDB	R109	R119	R135
PDC	R107	R121	R133
PDD	R105	R122	R136

Table 4: Test point, reference and functions on PCB 312-000-181 in the E-box.

Test Point	Reference	Function
TP6	VERT_DIF	In0, vertical deflection feedback signal
TP8	IN1B	LP filtered vertical deflection
TP9	IN1A	Lateral signal (LFM, torsion)
TP10	IN1C	Laser sum signal
TP15	HORZ_DIF	Horizontal deflection feedback signal
J20	VERT	Non-normalized vertical deflection
J35	HORZ	Non-normalized horizontal deflection



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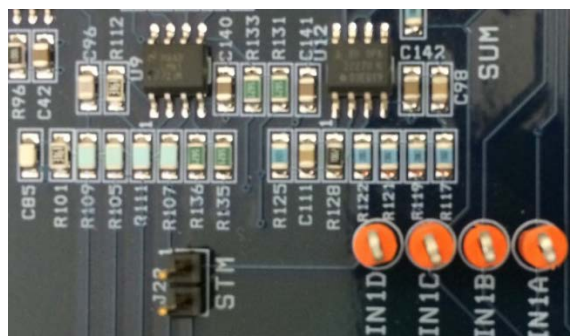


Figure 16. Test points for SUM and deflection signals. Refer to Table 3 for test point assignment.

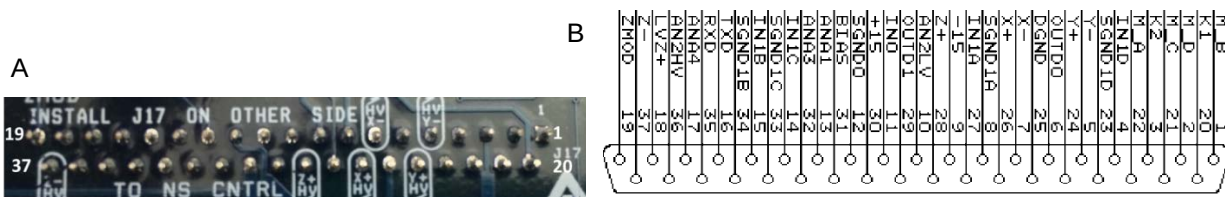


Figure 17. (A) The D37 controller-microscope cable connector at the back panel in the Ebox; (B) Pin out for the D37 connector. The +5V can be measured at K1 and K2 (Pin #3 and Pin #20)

For D3000, D3100 with blue PCB #312-000-005: Refer to “**For NSV**”.

Raw photodetector signals measured at J11: pin #9 (PDA), #11 (PDB), #27 (PDC), and #29 (PDD).

SUM and deflection can be tested on the left of the daughter board, see Figure 18.

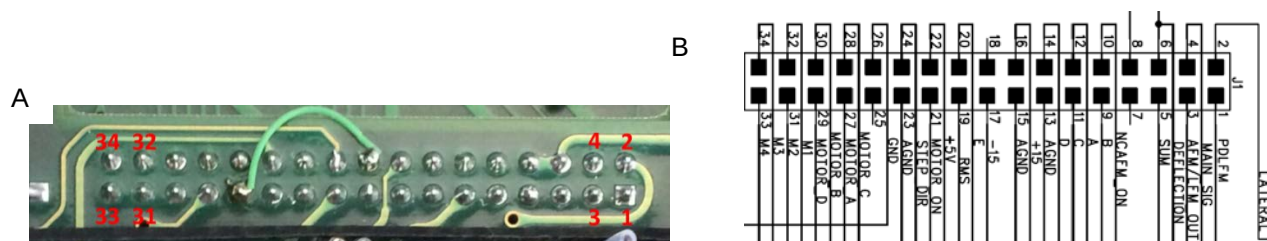


Figure 18. (A) Main connector to daughter board on the SSB; (B) Pin out for the connector.

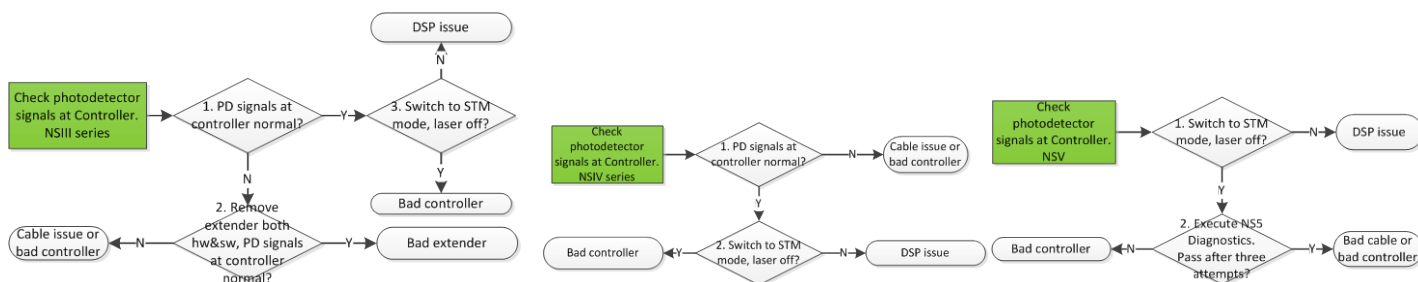
SUM and deflection signals leaving the back panel can be tested at the D37 controller-microscope cable connector, see Figure 19.



Figure 19. (A) The D37 controller-microscope cable connector at the back panel in the Ebox; (B) Pin out for the D37 connector.

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Section 6: Check photodetector signals at controller



For NSIII and NSIV series, if a Signal Access Module is available, there is no need to open the controller. Otherwise, this test requires opening controller for NSIII and NSIV series. After opening the controller, check that all cables are well-seated and no visible damages, such as burnt holes or mechanical failures.

6.1 For NSIII series:

Step 1: Test the IN0 (vertical deflection) signals on the Z board in the controller when the AFM software is operating. The IN0 test point seats on the Z board as shown in Figure 12. Perform the deflection/friction test. Refer to “NBSB0005_Deflection and Friction Test” and “NBSB0021_Deflection and Friction Test v7”

Step 2: This step is only for system with an extender. If no extender, skip this step. Remove the extender and change the software to the right configuration, connecting controller directly with the microscope. Then, repeat Step #1.

Step 3: This is to test if the DSP is functioning.

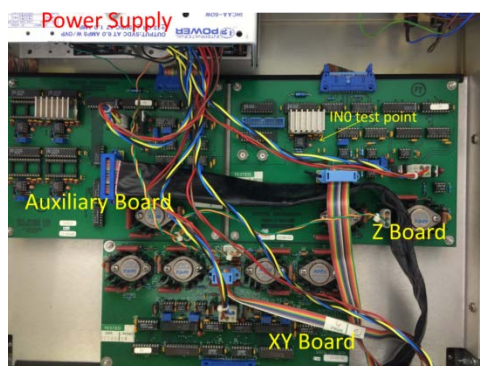


Figure 20. Architecture inside Nanoscope IIIA. The four primary components are labelled: Power supply assembly, auxiliary board, Z board, and XY board. The IN0 (vertical deflection) test point is indicated.

6.2 For NSIV series:

Step 1: Refer to Step #5 in Section 3, Figure 11, Step 1 in Section 6.1, and Figure 12.

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		Products: <u><i>D3000, 3100, D5000, and V</i></u>
Revision	Description	Date
A	Initial Release – (Teddy Huang and Pat McPhail)	12/23/2014
B	Rev B – (Teddy Huang and Pat McPhail)	03/30/2015

Step 2: This is to test if the DSP is functioning.

6.3 For NSV:

Step 1: This is to test if the DSP is functioning.

Step 2: Refer to “NBSB0003_Quick_Guide_for_the_NS5_Diagnostic_System_Tool” and Page 4-5 in “NBSB0146_Bulletin Erratic Scanner Movement”

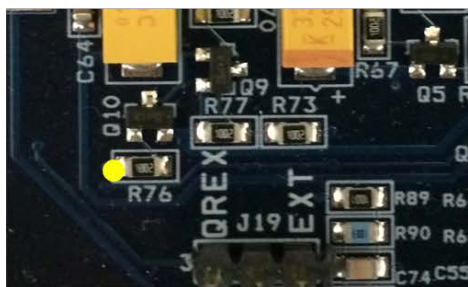
Appendix:

When the system is at STM mode, the STM logic voltage is set to high. This voltage can be measured at locations as shown in Figure 21. These PCB are the back panel in the E-box. Measurement requires open the E-box.

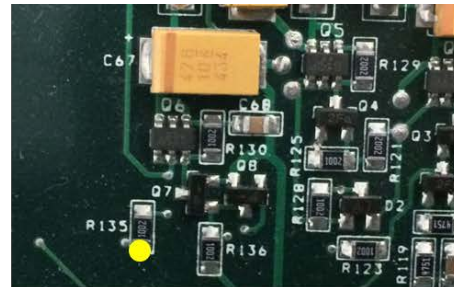
When the system is not at STM mode, the measurement should read 0 volt.



(A)



(B)



(C)

Figure 21. Position for measuring the STM logic high voltage. (A) PCB #312-000-005 for D3000 and D3100 systems, 5V measured at R51. This resistor is underneath the daughter board. Notice the voltage drop at different legs; (B) PCB #312-000-181 for D3000 and D3100 systems, 5V measured on the left leg of R76 as labelled; and (C) for Dimension V system, 3.3 V measured at the bottom leg of R135 as labelled.